Voronoi Homogeneity Analysis of Categorical Data

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TBD

Table of contents

Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All Rmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at https://github.com/deleeuw/voronoi

1 Introduction

2 Homogeneity Analysis

3 Voronoi Analysis

$$\sigma(X,Y,\Delta)=\sum_{i=1}^n\sum_{j=1}^m\|\delta_{ij}-d_{ij}(X,Y)\|^2$$

If $g_{i\ell}^j=1$ then $\{\delta_{ij}\}_\ell\leq\{\delta_{ij}\}_\nu$ for all $\nu=1,\cdots,k_j$.

$$\delta_{ij}'J\delta_{ij}=1$$

Majorization

$$A_{ij}:=\begin{bmatrix}e_ie_i'&-e_ie_j'\\-e_je_i'&e_je_j'\end{bmatrix}$$

$$Z:=\begin{bmatrix}X\\Y\end{bmatrix}$$

$$d_{ij}(X,Y)=\text{tr }Z'A_{ij}Z$$

$$V:=\sum_{i=1}^n\sum_{j=1}^m A_{ij}=\begin{bmatrix}mI&-ee'\\-ee'&nI\end{bmatrix}$$

If we define

$$V_1:=(n+m)^{-1}\begin{bmatrix}\frac{m}{n}E&-E\\-E&\frac{n}{m}E\end{bmatrix},\tag{1}$$

$$V_2:=\begin{bmatrix}J_n&0\\0&0\end{bmatrix},\tag{2}$$

$$V_3:=\begin{bmatrix}0&0\\0&J_m\end{bmatrix},\tag{3}$$

then

$$V=(n+m)V_1+mV_2+nV_3$$

Matrices V_1,V_2, and V_3 are symmetric projectors on orthogonal subspaces. It follows that V^+, the Moore-Penrose inverse3 of V, is

$$V^+=\frac{1}{n+m}V_1+\frac{1}{m}V_2+\frac{1}{n}V_3$$

$$B(X,Y):=\begin{bmatrix}B_{11}(X,Y)&B_{12}(X,Y)\\B_{21}(X,Y)&B_{22}(X,Y)\end{bmatrix}$$

Now $V_1B(X,Y)=0$ and thus

$$V^+B(X,Y)=\frac{1}{m}V_2B(X,Y)+\frac{1}{n}V_3B(X,Y)$$

4 The Cone

Consider the cone of all vectors x in $\mathbb{R}^n$ for which $x_i\leq x_1$ for all $i\neq 1$ and $e'x=0$. The extreme rays of the cone will be the $n-1$ vectors of the form $\alpha e_i+\beta(e-e_i)$ with $i\neq 1$. Thus element i is equal to α and all other elements (including x_1) are equal to beta. Thus we must have $\alpha>\beta$. The sum of the elements is $\alpha+(n-1)\beta$, which must be zero. Thus $\alpha=-(n-1)\beta$, which implies $\beta<0$ and $\alpha>0$. The extreme rays are positive multiples of the vectors $e_i-n^{-1}e$ with $i\neq 1$ and the cone is the set of all weighted sums with non-negative weights of these $n-1$ vectors. Here is a small example.

```
a <- -cbind(1, -diag(4))
b <- (diag(5)-(1 / 5))[, -1]
print(a)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]   -1    1    0    0    0
[2,]   -1    0    1    0    0
[3,]   -1    0    0    1    0
[4,]   -1    0    0    0    1
```

```
print(b)
```

```
     [,1] [,2] [,3] [,4]
[1,] -0.2 -0.2 -0.2 -0.2
[2,]  0.8 -0.2 -0.2 -0.2
[3,] -0.2  0.8 -0.2 -0.2
[4,] -0.2 -0.2  0.8 -0.2
[5,] -0.2 -0.2 -0.2  0.8
```

```
print(a %*% b)
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    0    0    0
[2,]    0    1    0    0
[3,]    0    0    1    0
[4,]    0    0    0    1
```

For later reference the sum of squares of the vectors $e_i - n^{-1}e$ is $(n-1)/n$, so the vectors $\sqrt{n/(n-1)}(e_i - n^{-1}e)$ are the vectors of length 1 on the extreme rays. The average of the extreme rays is

5 Normalized Monotone Regression

Suppose y is a vector with n elements. The problem we study is to minimixe the sum of squares $\sigma(x) := \|x-y\|^2$ over $x \in K$ and $x'Jx = 1$. Here

- K is the cone of vectors with $x_1 \leq x_2 \leq \cdots \leq x_n$,
- J is the centering matrix $J := I - n^{-1}ee'$, where
- e is the vector with all n elements equal to one.

Change variables to $\tilde{x} = Jx$. Then $x'Jx = \tilde{x}'\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|^2$ over β and $\tilde{x} \in K$, with $e'\tilde{x} = 0$ and $\tilde{x}'\tilde{x} = 1$. The minimizer for β is $\overline{y} := n^{-1}e'y$, and thus the problem becomes minimizing $\|\tilde{x} - Jy\|^2$ over $\tilde{x} \in K$, $\tilde{x}'\tilde{x} = 1$, and $e'\tilde{x} = 0$.

Forget about the constraint $e'\tilde{x} = 0$. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $M(Jy)/\|M(Jy)\|$, which automatically satisfies $e'\tilde{x} = 0$. Thus the solution of our original problem is

$$\hat{x} = \frac{M(y-\overline{y})}{\|M(y-\overline{y})\|} + \overline{y}.$$

$M(y-\overline{y}) = M(y) - \overline{y}$ and thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 + n\overline{y}^2 - 2\overline{y}e'M(y)$ nor $e'M(y) = e'y = n\overline{y}$. Thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 - n\overline{M(y)}^2$

$$\hat{x} = \frac{M(y) - \overline{y}e}{\|M(y-\overline{y})\|} + \overline{y}e = \frac{1}{\|M(y-\overline{y})\|}M(y) + \left\{1 - \frac{1}{\|M(y-\overline{y})\|}\right\}\overline{y}e$$

6 References